Distribution Shift in CNN-Based Seed Sorting: Causes, Detection, and Inference-Time Correction

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Abstract

CNN-based machine vision systems are widely used for grain quality inspection, but their performance degrades when imaging conditions deviate from those seen during training. The effect of such imaging-induced distribution shift on threshold-based sorting decisions has, to our knowledge, not been previously studied. This paper investigates whether imaging deviations deform the yield-threshold curve, whether such deformations can be detected, and whether inference-time correction can restore the curve without retraining. Experiments are conducted on GrainSet-tiny, a subset of the GrainSet wheat kernel database, using a ResNet-50 classifier. Results show that Gaussian noise causes the most severe deformation (ΔAUC-YT $= -0.423$ at severity 5), that both AUC-YT deviation and Wasserstein distance reliably detect deformation from severity 3 onward, and that BatchNorm adaptation achieves a mean AUC-YT recovery of 31.9% without retraining. These findings demonstrate that imaging-induced distribution shift is a measurable and partially correctable cause of yield deviation in deployed seed sorting systems.

1. Introduction

Grain sorting at an industrial scale relies on CNN-based machine vision systems to classify kernels as good or bad at high throughput. These models are typically trained on images captured by the production machine on which they are later deployed, so the relevant failure mode is not a mismatch between training and deployment hardware but a change in imaging conditions on the same machine over time. The classifier outputs a confidence score in $[0, 1]$, and an operator sets a threshold above which a seed is directed to the accept fraction. This threshold is typically set once per seed lot, under the assumption that the ratio of good to bad seeds, and therefore the proportion of seeds whose confidence score exceeds the threshold (the *yield*), remains stable for the duration of the lot. Under this assumption, the operator can rely on a predictable accept fraction and a consistent confidence distribution throughout sorting.

In practice, operators observe that the yield at a fixed threshold deviates from the expected value, and adjust the threshold repeatedly to maintain a target yield. One plausible cause is that imaging conditions on the production machine deviate from those present during training, due to lighting variation, lens contamination, or camera replacement. Since the model was optimised for a specific image distribution, such deviations cause confidence scores to shift systematically, affecting the proportion of seeds that meet the acceptance threshold. The practical consequence is that yield deviates from the operator's target, causing either over-rejection of good seeds (economic loss) or under-rejection of damaged seeds (quality failure) (Ülger & Sürer, 2024).

Distribution shift is the mismatch between training and test data distributions and is a well-recognised challenge in deployed machine learning systems (Hendrycks & Dietterich, 2019; Ülger & Sürer, 2024). In the context of imaging systems, such shifts arise from changes in sensor characteristics, lighting conditions, or optical properties. Test-time adaptation (TTA) methods (Liang et al., 2024) have been proposed to address this without retraining, including BatchNorm statistics re-estimation (Schneider et al., 2020; Li et al., 2017) and entropy minimisation (Wang et al., 2021). However, these methods are typically evaluated on standard benchmarks such as ImageNet-C (Hendrycks & Dietterich, 2019), and their effectiveness in industrial sorting applications where decisions are threshold-based and yield is the primary metric has, to our knowledge, not been studied. It remains unclear whether such methods can restore the expected yield at a given threshold when imaging conditions deviate from those seen during training.

This paper investigates whether imaging deviations cause measurable deformations of the yield-threshold curve, whether such deformations can be automatically detected, and whether inference-time interventions can restore the curve without retraining. Experiments are conducted on GrainSet (Fan et al., 2023a), a large-scale database of single-kernel wheat images with expert annotations. A ResNet-50 classifier (He et al., 2016) is trained on the clean training set and evaluated against test sets corrupted with three types of imaging deviations: brightness scaling, Gaussian blur, and Gaussian noise, each at five severity levels. Detection is assessed using the area under the yield-threshold curve (AUC-YT) and the Wasserstein distance between confidence score distributions. Three inference-time correction methods are evaluated: per-channel input normalisation, BatchNorm statistics re-estimation (Schneider et al., 2020; Li et al., 2017), and test-time entropy minimisation (Wang et al., 2021). The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 describes the methods. Section 4 presents the experimental results. Sections 5 and 6 provide discussion and conclusions. This paper addresses the following research questions:

1. Can imaging deviations cause deformations of the yield-threshold curve, and if so, what is the impact of different deviation types and severities?
2. Can such deformations be automatically detected through divergences in the confidence score distribution?
3. Can inference-time model interventions restore the yield-threshold curve without retraining, and if so, which interventions are effective?

2. Related work

Distribution shift is a fundamental challenge in machine learning: models trained on one distribution often fail when test conditions differ (Hendrycks & Dietterich, 2019; Liang et al., 2024). A common form is covariate shift, where the input distribution changes while the underlying labelling function remains unchanged. Hendrycks and Dietterich (Hendrycks & Dietterich, 2019) introduced ImageNet-C as a systematic benchmark for evaluating robustness to common image corruptions including noise, blur, and brightness shifts, designed to simulate real-world imaging degradations. Their findings showed that CNNs are surprisingly sensitive to even mild corruptions, motivating a line of research into test-time adaptation methods that can correct for such shifts without retraining.

A key component exploited by several TTA methods is Batch Normalisation (BN) (Ioffe & Szegedy, 2015), which normalises activations using statistics computed during training. When test distribution shifts, these frozen statistics become mismatched. Li et al. (Li et al., 2017) showed through their AdaBN approach that simply replacing training BN statistics with test-set statistics significantly improves performance under domain shift. Schneider et al. (Schneider et al., 2020) extended this to corruption robustness, showing BN re-estimation is effective against the types of corruptions in ImageNet-C. Wang et al. (Wang et al., 2021) went further by also optimising the BN affine parameters (scale and shift) via entropy minimisation, achieving strong adaptation with minimal compute overhead. Rabanser et al. (Rabanser et al., 2019) systematically evaluated methods for detecting dataset shift, finding that statistical tests on learned representations outperform tests on raw inputs. Among the metrics evaluated, the Wasserstein distance proved particularly sensitive to distributional divergence, capturing both location and shape changes. In the context of seed sorting, detecting such divergence is practically important: an operator needs to know whether observed yield deviations are caused by imaging deviations before applying any correction.

Machine vision for grain quality inspection is an active research area, with deep learning demonstrating strong performance on grain defect classification (Ülger & Sürer, 2024; Fan et al., 2023b). Fan et al. (Fan et al., 2023a) introduced GrainSet, a large-scale database of over 350,000 single-kernel images across four cereal grain types with expert annotations for eight damage categories. A preview subset, GrainSet-tiny, containing approximately 2% of the full dataset, is also publicly available and is used in this work. The same authors also published work on identifying defective grains using deep learning (Fan et al., 2023b) and on domain-adaptive recognition across grain types (Fan et al., 2022). However, none of these works consider the effect of imaging deviations on sorting decisions, nor do they evaluate inference-time correction methods in this context. This paper addresses that gap.

3. Methods

3.1. Dataset

Due to computational constraints, experiments are conducted on GrainSet-tiny rather than the full wheat dataset. GrainSet-tiny contains 2,000 wheat kernel images sampled uniformly from the full dataset, while preserving the original class distribution of 59.9% normal kernels and 40.1% damaged or unsound kernels. The dataset is split into training, validation, and test sets using a stratified 70/15/15 split, yielding 1,400 training, 300 validation, and 300 test samples. The class distribution is preserved across all splits. We acknowledge that results obtained on GrainSet-tiny may not fully generalise to the full dataset, and discuss this limitation in Section 5. Inverse-frequency class weights are applied during training to account for the class imbalance.

3.2. Model

ResNet-50 (He et al., 2016) is selected as the backbone as it reflects the architecture commonly used in industrial seed sorting applications and serves as the benchmark classifier in the GrainSet paper (Fan et al., 2023a), enabling direct comparison with reported baselines. The final fully-connected layer is replaced by a two-class linear head. Input images are resized to 224×224 pixels and normalised using

ImageNet mean and standard deviation prior to training.

The model is trained for up to 50 epochs using stochastic gradient descent (SGD) with a learning rate of 0.0012, momentum of 0.9, and weight decay of 10^{-4}. Step-decay milestones at epochs 20 and 40 (factor 0.1) are configured but, in practice, the model converges within the first few epochs on the 1,400-sample training set and these milestones are never reached. A batch size of 32 is used with cross-entropy loss weighted by inverse class frequency, computed as $w_c = \frac{N}{2 \cdot N_c}$, where N is the total number of training samples and N_c is the number of samples in class c. The checkpoint achieving the highest validation F1 score is retained for all subsequent experiments; on GrainSet-tiny this is reached at epoch 5 with a validation F1 of 0.986, after which validation F1 plateaus.

3.3. Imaging Deviations

Three types of imaging deviation are simulated, each motivated by a real-world cause observed in industrial sorting environments. Brightness scaling multiplies all pixel values by a scalar factor, simulating lighting drift or LED ageing. Gaussian blur applies a low-pass filter with a given standard deviation, simulating lens contamination, sensor ageing, or focus drift. Gaussian noise adds zero-mean noise with a given standard deviation, simulating differences in sensor noise levels between cameras or due to sensor degradation.

Each deviation type is applied at five severity levels, spanning mild to severe degradation. The exact parameter values are listed in Table 1. In addition to the three individual deviation types, all four pairwise and full combinations are evaluated: brightness+blur, brightness+noise, blur+noise, and all three combined. This yields 35 deviated test conditions in total (7 deviation types $\times$ 5 severity levels). All corruptions are applied at inference time only. The model is never retrained on corrupted images.

Table 1. Corruption severity parameters. Brightness factor < 1 darkens the image; > 1 brightens it. Blur σ is in pixels on a 224×224 image. Noise standard deviation is in normalised $[0, 1]$ pixel space.

Severity	Brightness factor	Blur σ	Noise std
1	0.60	0.5	0.02
2	0.75	1.0	0.05
3	0.90	2.0	0.10
4	1.25	3.0	0.15
5	1.50	4.5	0.20

Figure 1 illustrates example wheat kernel images under each corruption type and severity level.

3.4. Evaluation Metrics

The yield-threshold curve plots the fraction of seeds accepted by the model as a function of the confidence threshold $t \in [0, 1]$. At threshold t, a seed is accepted if its confidence score exceeds t, so the yield at threshold t is the proportion of test samples with a confidence score of at least t – equivalently, the empirical survival function of the model's confidence scores on the test set. Under clean imaging conditions this curve has a characteristic S-shape, reflecting the bimodal distribution of confidence scores produced by a well-trained binary classifier.

The area under the yield-threshold curve (AUC-YT) is used as the primary scalar summary of curve shape:

$$\text{AUC-YT} = \int_0^1 \text{yield}(t)\, dt \tag{1}$$

approximated via the trapezoidal rule over 1,000 evenly spaced threshold values. Because $\text{yield}(t)$ is the survival function of the confidence scores, AUC-YT equals the mean confidence score on the test set, and is distinct from AUC-ROC, which integrates true-positive rate against false-positive rate. For a well-calibrated binary model this implies that the clean-baseline AUC-YT should approximately equal the proportion of 'good' kernels in the test set: the observed value of 0.597 is consistent with the 59.9% normal-class proportion of GrainSet-tiny. The deviation from the clean baseline is reported as $\Delta\text{AUC-YT} = \text{AUC-YT}_{\text{deviated}} - \text{AUC-YT}_{\text{baseline}}$, where negative values indicate that the deviated curve sits below the clean curve over most of the threshold range.

As a complementary detection metric, the Wasserstein distance (Rabanser et al., 2019) is computed between the baseline confidence score distribution and the deviated confidence score distribution. Unlike AUC-YT, which summarises the curve shape, the Wasserstein distance directly measures the distributional divergence in the raw confidence scores, capturing both shifts in location and changes in shape.

3.5. Inference-Time Correction Methods

Three inference-time correction methods are evaluated, each applied to the frozen trained model without any retraining.

Per-Channel input normalisation normalises each test batch to zero mean and unit standard deviation per channel, estimated from the batch itself. This corrects for global brightness and contrast shifts by replacing the fixed ImageNet normalisation statistics with statistics derived from the test distribution. The normalised batch is subsequently re-scaled to the ImageNet distribution before being passed to the model.

BatchNorm adaptation (BN adapt) switches all Batch-

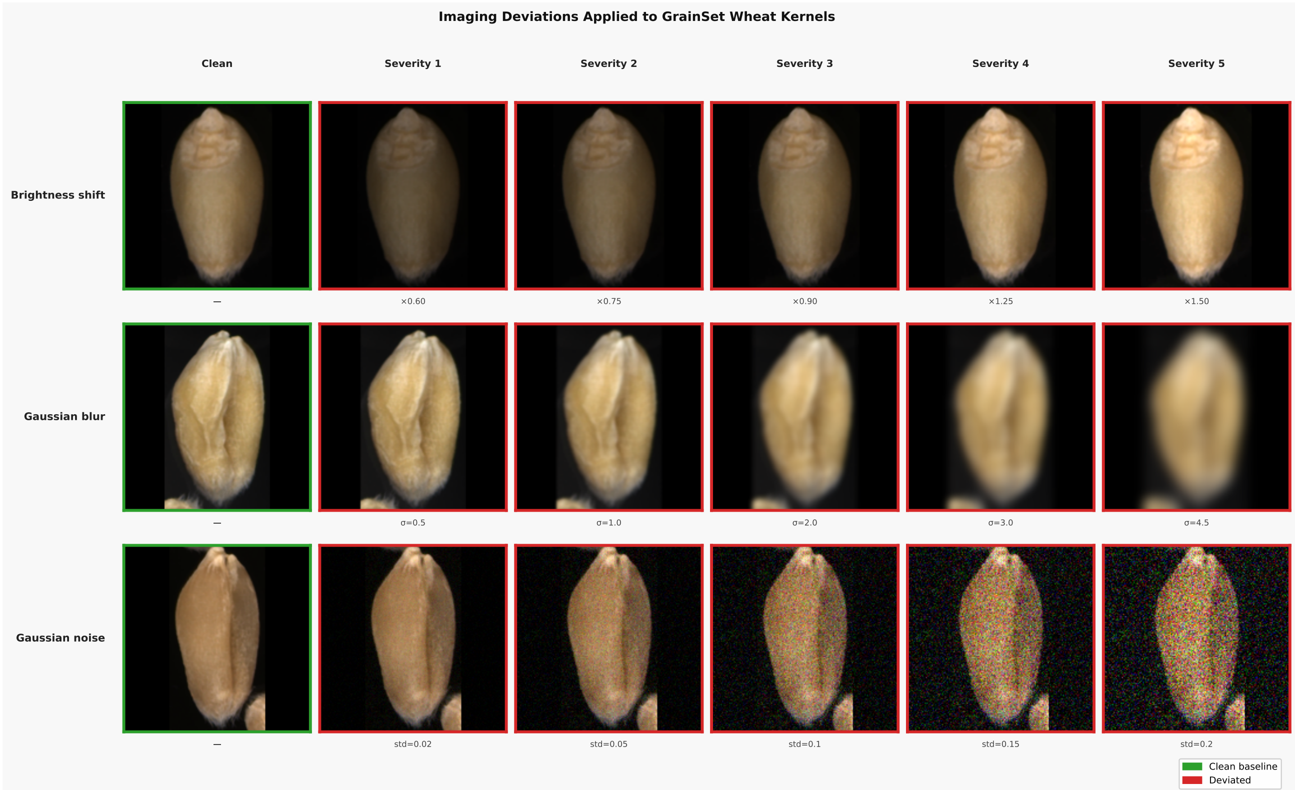

Figure 1. Example wheat kernel images under each imaging deviation type (rows) and severity level (columns). Left column shows the clean baseline image. Severity increases left to right.

Norm layers to training mode during a warm-up pass over the full test set, allowing the running mean and variance to be re-estimated from the deviated test distribution using a cumulative moving average. The model is then switched back to evaluation mode and inference is performed using the adapted statistics. This approach follows the AdaBN protocol of Li et al. (Li et al., 2017) and the corruption robustness extension of Schneider et al. (Schneider et al., 2020).

Test-time entropy minimisation (TENT) (Wang et al., 2021) extends BN adaptation by additionally optimising the BatchNorm affine parameters scale (γ) and shift (β) via one gradient step per batch to minimise the Shannon entropy of the model's predictions. Minimising entropy encourages the model to make confident predictions on the test batch, effectively adapting the decision boundary to the deviated distribution. All other model parameters remain frozen throughout.

4. Experimental Results

4.1. Implementation Details

Experiments are implemented in PyTorch 2.11 (Paszke et al., 2019) with torchvision. All experiments are run on CPU due to hardware constraints. A fixed random seed of 42 is used throughout for reproducibility. Code is publicly available at `https://github.com/gbudding/cnn-distribution-shift`.

4.2. H1: Yield-Threshold Curve Deformation

Table 2 and Figure 2 report the effect of each imaging deviation on the yield-threshold curve. All three deviation types cause measurable deformation at higher severities, confirming H1.

Gaussian noise causes the most severe deformation, with ΔAUC-YT $= -0.423$ at severity 5, reducing the AUC-YT from a baseline of 0.597 to 0.175. Gaussian blur is the second most damaging deviation, reaching ΔAUC-YT $= -0.136$ at severity 5. Brightness scaling has the smallest effect, with a maximum ΔAUC-YT $= -0.085$ at severity 5.

Table 2. AUC-YT and ΔAUC-YT for individual corruptions at all severity levels. Baseline AUC-YT = 0.597.

Type	Sev	AUC-YT	ΔAUC-YT
Brightness	1	0.584	-0.013
	3	0.599	+0.002
	5	0.513	-0.085
Blur	1	0.599	+0.001
	3	0.518	-0.079
	5	0.461	-0.136
Noise	1	0.613	+0.016
	3	0.410	-0.187
	5	0.175	-0.423

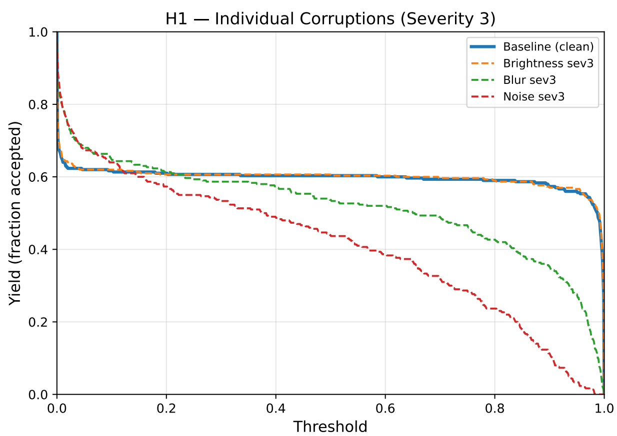

Figure 2. Yield-threshold curves for individual corruption types at severity 3, compared against the clean baseline (AUC-YT = 0.597).

Combination corruptions amplify the deformation beyond what individual corruptions produce. The brightness and noise combination reaches ΔAUC-YT $= -0.548$ at severity level 5, the largest deformation observed. All combinations involving noise produce substantially larger deformations than blur or brightness alone, confirming the dominant role of sensor noise in yield-threshold curve deformation.

An unexpected finding is that mild corruptions at severity 1 occasionally produce slight positive ΔAUC-YT. For example, Gaussian noise at severity 1 yields ΔAUC-YT $=$ $+0.016$. This may indicate a mild implicit-regularisation effect of low-amplitude noise on borderline confidence scores, but with only 300 test samples per condition, sampling noise of this magnitude cannot be ruled out. The effect disappears at severity 2 and above.

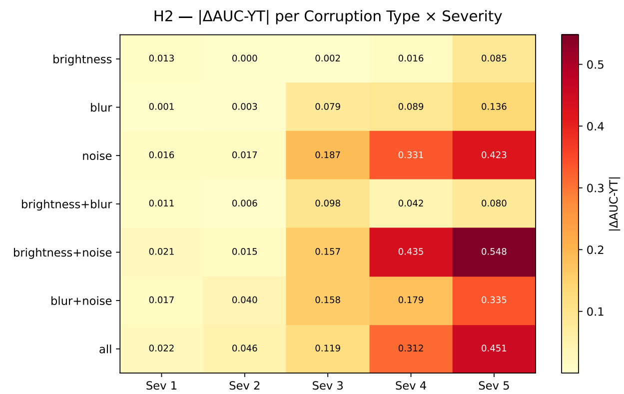

Figure 3. $|\Delta$AUC-YT$|$ for all corruption types and severity levels. Both AUC-YT deviation and Wasserstein distance (Appendix A) show consistent patterns, with noise producing the largest detectable deformation.

4.3. H2: Detection of Yield-Threshold Curve Deformation

Figure 3 shows the $|\Delta$AUC-YT$|$ for all 35 deviated conditions. Both detection metrics follow consistent patterns across all corruption types and severity levels. Noise is the most detectable deviation type, with $|\Delta$AUC-YT$| = 0.423$ and a Wasserstein distance of 0.467 at severity 5. Blur is clearly detectable from severity 3 onward, while brightness shifts are hardest to detect, with both metrics remaining below 0.090 across all severities.

The two metrics are strongly correlated across all 35 conditions, as shown in Figure 5 in the appendix. This confirms that AUC-YT and Wasserstein distance capture the same underlying distributional divergence, and that either metric could serve as a reliable indicator of imaging-induced yield deviation.

Statistical testing using one-sample t-tests against the null hypothesis of no mean deviation ($\mu_{\Delta\text{AUC-YT}} = 0$) yields no significant results across any corruption type (all $p > 0.05$). This is expected given the small number of severity levels ($n = 5$) available for testing, which limits statistical power. A Spearman rank correlation between severity and $|\Delta$AUC-YT$|$ provides a more appropriate test of monotone trend. The correlation is perfect and statistically significant for noise ($\rho = 1.0$, $p = 0.017$) and blur ($\rho = 1.0$, $p = 0.017$), but weaker and non-significant for brightness ($\rho = 0.7$, $p = 0.19$), reflecting the non-monotone pattern at low brightness severities where a moderate darkening (factor 0.60, severity 1) produces larger deformation than milder deviations at severities 2–3. H2 is therefore partially supported: deformation is detectable through both metrics, most clearly for noise and blur at severity 3 and above.

4.4. H3: Inference-Time Restoration of the Yield-Threshold Curve

Three inference-time correction methods are evaluated across all 35 deviated conditions. Recovery is defined as the proportion of the AUC-YT deviation that the correction removes:

$$\text{Recovery} = \frac{\text{AUC-YT}_{\text{corrected}} - \text{AUC-YT}_{\text{deviated}}}{\text{AUC-YT}_{\text{baseline}} - \text{AUC-YT}_{\text{deviated}}} \quad (2)$$

A recovery of 100% restores the deviated AUC-YT exactly to the clean baseline; 0% indicates no improvement; negative values indicate that the corrected curve is further from the baseline than the uncorrected one. The denominator is small for mild deviations, so per-condition recovery percentages can be numerically unstable at low severities and raw AUC-YT values are reported where more informative. The full recovery heatmap is provided in Appendix A (Figure 6).

Per-channel input normalisation fails to restore the yield-threshold curve and in most conditions makes it substantially worse. The mean corrected AUC-YT across the 35 conditions is 0.113, compared to a mean uncorrected AUC-YT of 0.474 and a clean baseline of 0.597: the method moves the curve dramatically further from the baseline rather than towards it. In 33 of the 35 conditions the corrected AUC-YT is further from the baseline than the uncorrected one; the two exceptions are the most severely deviated conditions (brightness+noise and all-combined at severity 5), where the uncorrected curve has already collapsed to near zero and there is little left to make worse. Rather than correcting for imaging deviations, per-channel normalisation destroys the discriminative information in the input by removing the texture and contrast cues that the model relies on for classification, suggesting it is incompatible with texture-dependent inspection tasks such as grain quality classification.

BatchNorm adaptation achieves a mean AUC-YT recovery of +31.9% across all conditions, making it the most effective correction method evaluated. It is particularly effective for brightness and blur deviations, restoring AUC-YT to values close to the clean baseline (e.g. blur severity 5: from 0.461 to 0.587). For noise deviations, recovery is partial but consistent, with AUC-YT restored from 0.175 to 0.552 at severity 5. The mean Wasserstein distance to the baseline is reduced from 0.163 (no correction) to 0.046 after BN adaptation, confirming that the confidence score distribution is substantially realigned with the clean baseline.

TENT achieves a mean AUC-YT recovery of +19.7%, somewhat lower than BN adaptation overall. However, TENT outperforms BN adaptation at higher blur severities, reaching AUC-YT = 0.612 at blur severity 5 compared to 0.587 for BN adaptation. For severe noise and noise combination corruptions, TENT performs slightly below BN adaptation, suggesting that entropy minimisation may over-adapt when image features are severely degraded. The mean Wasserstein distance after TENT (0.028) is lower than after BN adaptation (0.046), indicating that TENT brings the confidence score distribution closer to the baseline despite achieving lower mean AUC-YT recovery.

H3 is partially supported. BN adaptation and TENT both partially restore the yield-threshold curve without retraining, with BN adaptation being more robust across all deviation types. Per-channel input normalisation is ineffective for this task.

5. Discussion

The results confirm that imaging deviations systematically deform the yield-threshold curve, that this deformation is detectable through distributional metrics, and that inference-time correction can partially restore the curve. Each finding has practical implications for the deployment of CNN-based seed sorting systems.

Gaussian noise emerges as the most damaging deviation type, producing AUC-YT reductions of up to 0.423 at severity 5. This is consistent with the nature of the classification task: grain quality inspection relies heavily on fine surface texture to distinguish normal kernels from damaged ones. Gaussian noise corrupts precisely these high-frequency texture features, while brightness scaling and Gaussian blur preserve the overall kernel shape and colour information that the model also uses. This suggests that sensor noise control is the most critical imaging parameter to monitor in deployed sorting systems.

The failure of per-channel input normalisation is informative. By normalising each batch to zero mean and unit variance per channel, this method removes the global brightness and contrast information that distinguishes deviated images from clean ones, but simultaneously suppresses the local texture contrast that the model relies on for classification. This result highlights a fundamental tension in input normalisation: corrections designed to remove unwanted variations can also remove discriminative signal when the two are correlated. BN adaptation avoids this problem by operating on internal feature distributions rather than raw pixel values, allowing it to correct for distributional shift without destroying input information. TENT's slightly lower mean recovery compared to BN adaptation, despite achieving a lower Wasserstein distance, suggests that entropy minimisation may over-adapt under severe noise conditions, optimising for confident but potentially incorrect predictions when image features are severely degraded.

Several limitations should be noted. First, experiments are conducted on GrainSet-tiny, which contains only 2,000 samples. Results on the full wheat dataset of 200,000 images may differ, particularly with respect to model calibration and the absolute magnitude of AUC-YT values. Second, the imaging deviations evaluated are synthetic approximations of real-world degradations. Actual camera drift, lens contamination, and sensor ageing may produce more complex and correlated deviations than the independent corruptions studied here. Third, only wheat kernels are studied: generalisation to maize, sorghum, and rice cannot be assumed without further experimentation. Finally, statistical testing is limited by the small number of severity levels ($n = 5$), restricting the power of significance tests. A further limitation is that the correction methods evaluated operate offline: BN adaptation requires a warm-up pass over the full test set before inference begins, and TENT updates per batch rather than per sample. Neither method is suitable for continuous online adaptation during active sorting, where the model would need to adapt incrementally as imaging conditions drift over the course of a production run.

Future work should address these limitations by training on the full wheat dataset and evaluating on real deviated images captured in production environments. Extending the analysis to other grain species would test the generalisability of the findings. A particularly promising direction is the development of online adaptation methods that update model statistics continuously during sorting, without requiring a separate calibration phase or interrupting production throughput. Finally, investigating whether more recent backbone architectures such as ConvNeXt (Liu et al., 2022) or vision transformers (Dosovitskiy et al., 2021) show different robustness profiles under imaging deviations would help determine whether the findings generalise across backbone choices.

6. Conclusions

This paper investigated the effect of imaging deviations on the yield-threshold curve of a CNN-based seed sorting system, and evaluated methods for detecting and correcting such deviations at inference time. Three research questions were addressed using a ResNet-50 classifier trained on GrainSet-tiny wheat kernel images.

With respect to Q1, all three imaging deviation types cause measurable deformation of the yield-threshold curve, with Gaussian noise producing the most severe effect (ΔAUC-YT $= -0.423$ at severity 5) and combination corruptions amplifying deformation further. With respect to Q2, both AUC-YT deviation and Wasserstein distance reliably track the degree of deformation, with noise and blur detectable from severity 3 onward. With respect to Q3, BatchNorm adaptation achieves a mean AUC-YT recovery of 31.9% without retraining, making it the most practical inference-time correction method evaluated. TENT achieves 19.7% mean recovery and outperforms BN adaptation at higher blur severities, while per-channel input normalisation fails entirely due to the suppression of discriminative texture features.

These findings demonstrate that imaging-induced distribution shift is a measurable and partially correctable cause of yield deviation in CNN-based seed sorting, and that BatchNorm adaptation is a promising lightweight intervention for deployed systems. Future work should validate these findings on the full GrainSet wheat dataset and on real imaging deviations captured in production environments. The development of online adaptation methods suitable for continuous use during active sorting represents a particularly promising direction for follow-up research.

A. Additional Figures

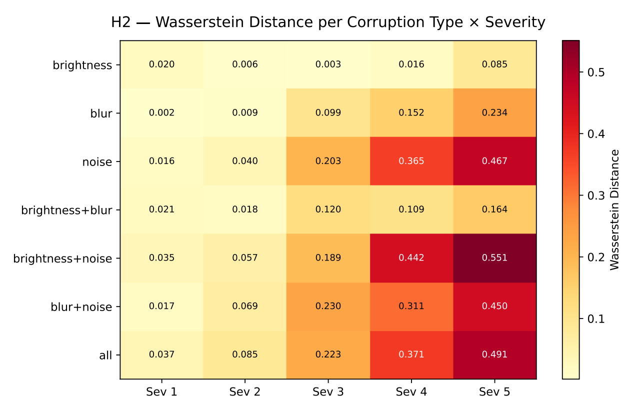

Figure 4. Wasserstein distance between baseline and deviated confidence score distributions per corruption type and severity level.

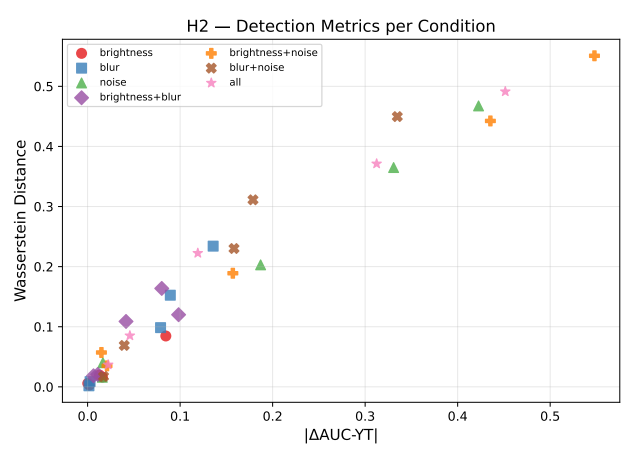

Figure 5. Wasserstein distance versus $|\Delta$AUC-YT$|$ for all 35 deviated conditions, coloured by corruption type. The strong positive correlation confirms that both metrics detect the same underlying distributional divergence.

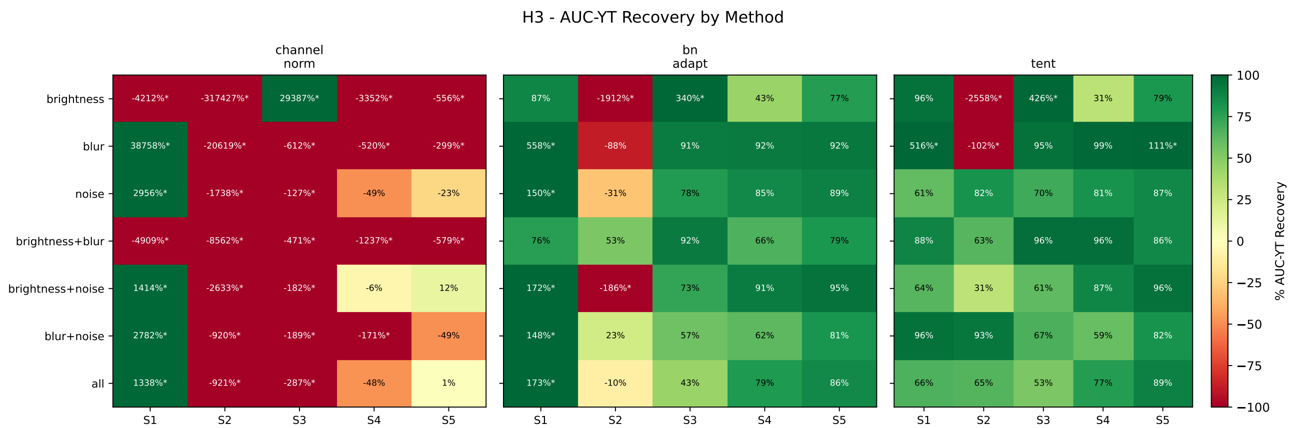

Figure 6. AUC-YT recovery percentage per correction method, corruption type, and severity level. The `no_correction` method is omitted (recovery is 0% by construction). Colour scale is clipped at $\pm 100\%$; cells marked with an asterisk fall outside this range but retain their numerical value.

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